EfficientNet-B3-Based Classification of Fetal Ultrasound Planes with Transfer Learning and Comparative Analysis

ABSTRACT:

Accurate identification of fetal ultrasound planes is essential for diagnosis but remains operator-dependent. This work presents a fine-tuned **EfficientNet-B3** model for six-class classification on the Fetal Planes DB. Using transfer learning with progressive unfreezing, label smoothing, and augmentation, the model achieves **93.15% test accuracy**. A comparative evaluation with **MobileNetV3** demonstrates the superior performance of EfficientNet-B3, particularly in terms of class-wise balance and robustness.

Analysis via confusion matrix and Grad-CAM highlights challenges in the **"Other"** and **"Fetal thorax"** classes. While results demonstrate strong performance, limitations such as **label noise, image-level splitting, and limited explainability** remain for clinical deployment.

1. INTRODUCTION:

Standardized fetal ultrasound planes are essential for prenatal assessment, enabling measurement of key anatomical structures for growth monitoring and anomaly detection. However, mid-trimester screening involves reviewing multiple images, making manual plane selection time-consuming and prone to variability. Recent advances in deep CNNs have demonstrated expert-level performance, highlighting their potential for automation.

Burgos-Artizzu et al. introduced the Fetal Planes DB, comprising 12,400 labeled ultrasound images across six classes. Building on this dataset, this work fine-tunes **EfficientNet-B3** using a modern PyTorch pipeline to evaluate classification performance and identify remaining challenges in fetal plane recognition.

2. MODEL ARCHITECTURE:

Proposed EfficientNet-B3 Architecture

The proposed model is based on **EfficientNet-B3**, which uses a compound scaling strategy to balance network depth, width, and resolution for improved efficiency and performance. The backbone is initialized with **ImageNet-1K pretrained weights** and operates on input images of size **260 × 260 pixels**.

Feature extraction is performed using stacked **MBConv blocks with squeeze-and-excitation**, enabling efficient representation learning. The default classifier is replaced with a **custom head**:
Dropout(0.4) → Linear(1536 → 512) → SiLU → Dropout(0.2) → Linear(512 → 6)

Training follows a **two-stage fine-tuning strategy**, where the classifier head is trained first (~12.2M parameters), followed by unfreezing the top MBConv blocks (~15.3M parameters) for domain adaptation.

The model is optimized using **AdamW (lr = 1e-4, weight_decay = 1e-4)** with **label-smoothed cross-entropy (ε = 0.1)**. A **3-epoch warm-up** followed by **cosine annealing** is used for stable convergence. **Mixed precision training** (torch.cuda.amp.autocast) is applied to improve computational efficiency.

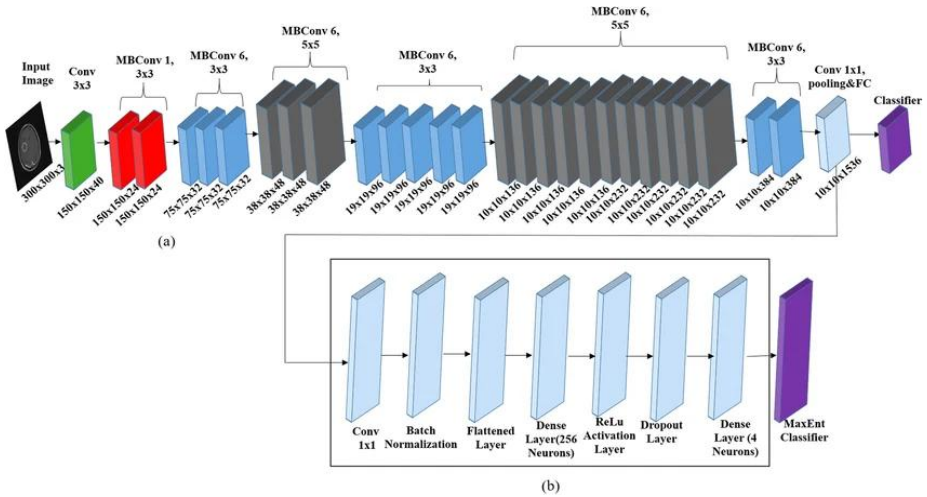

Fig. 1 EfficientNet-B3 model architecture with custom classification head.

3. RESEARCH GAP:

Work / Citation	Task & Dataset	Method / Architecture	Key Findings	Relevance to This Work & Gap
Burgos-Artizzu et al., Scientific Reports 2020 – Fetal Planes DB s41598-020-67076-5.pdf	Six-class maternal–fetal plane classification on 12,400 2D images (abdomen, brain, femur, thorax, cervix, Other).	Benchmarks many CNNs (VGG, Inception, ResNet, DenseNet) trained from ImageNet and fully fine-tuned; best model DenseNet-169.s41598-020-67076-5.pdf	Shows CNNs can match trained research technicians; best top-1 error ≈ 6.2% (≈93.8% accuracy) with DenseNet-169; highlights issues with class imbalance, fine-grained brain planes, and "Other" class.s41598-020-67076-5.pdf	Provides the core dataset and DenseNet-based baselines; this project instead uses EfficientNet-B3 with a modern training recipe (progressive unfreezing, label smoothing, strong augmentation) to show that a single, compact backbone can reach 93.15% accuracy on the same six-class task while remaining simpler than later ensembles
Baumgartner et al., SonoNet 2017 – Real-time plane detection arxiv+1	Real-time detection and localisation of 13 fetal standard scan planes in freehand 2D ultrasound videos.	VGG-style CNN modified to fully convolutional "SonoNet" with weak supervision and bounding-box localisation.arxiv+1	Achieves average F1 ≈ 0.80 for real-time detection, 90%+ accuracy for retrospective frame retrieval, and reasonable localisation accuracy.arxiv+1	Focuses on online detection and localisation from video streams, not static image classification; motivates automatic standard-plane recognition but leaves open a clean, EfficientNet-based image-only classifier on Fetal Planes DB, which this work provides using EfficientNet-B3.
Sivasubramanian et al., 2024 – Efficient CNN-attention FP arxiv+2	Fetal plane classification on the largest benchmark ultrasound dataset (Fetal Planes DB).	Lightweight EfficientNet-B0 / EfficientNetV2B0 backbones + attention modules + 3-layer MLP classifier; Grad-CAM for explainability.arxiv+2	Reports Top-1 accuracy ≈ 96.25% and F1 ≈ 0.958 with ~40× fewer parameters than some transformer and ensemble baselines; deployable on edge devices.arxiv+2	Shows that EfficientNet backbones combined with attention are highly effective for fetal plane classification; this project focuses instead on a backbone-only EfficientNet-B3 baseline under a fixed pipeline, establishing how far a single well-tuned EfficientNet model can go (93.15% test accuracy) before adding extra attention modules or multi-stage architectures

4. METHODOLOGY:

Progressive Unfreezing Strategy

A two-stage fine-tuning protocol was adopted to mitigate catastrophic forgetting. In Stage 1 (epochs 1–4), only the custom classifier head was trained while the backbone remained frozen, enabling the model to learn task-specific features. In Stage 2 (epoch 5 onward), the top three MBConv blocks of the backbone were unfrozen and trained with a reduced learning rate (1e-5), allowing gradual adaptation of deeper feature representations to ultrasound imagery.

5. DATASET, PREPROCESSING & AUGMENTATION:

5.1 Dataset Overview

The Fetal Planes DB contains 12,400 ultrasound images collected from 1,792 routine clinical acquisitions. Each image is annotated with a fetal plane label and quality score. Six plane classes are present, with a pronounced imbalance favouring Fetal brain and Fetal femur classes.

Class	Count	Notes
Fetal brain	3,994	Largest class; high within-class variation (transcerebellar, transventricular, transthalamic planes)
Fetal femur	2,652	High image quality; relatively homogeneous appearance
Other	1,040	Mixed anatomy; noisiest class — includes mislabelled frames
Fetal abdomen	944	Second most imbalanced; abdominal circumference standard plane
Fetal thorax	750	Smallest standard-plane class; cardiac structures visible
Maternal cervix	648	Structurally distinct from fetal planes; trans-labial acquisition

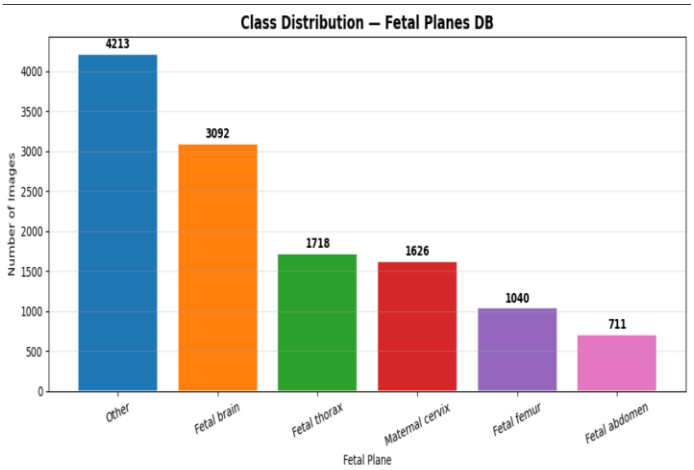

Fig. 2 Class distribution of the Fetal Planes DB data.

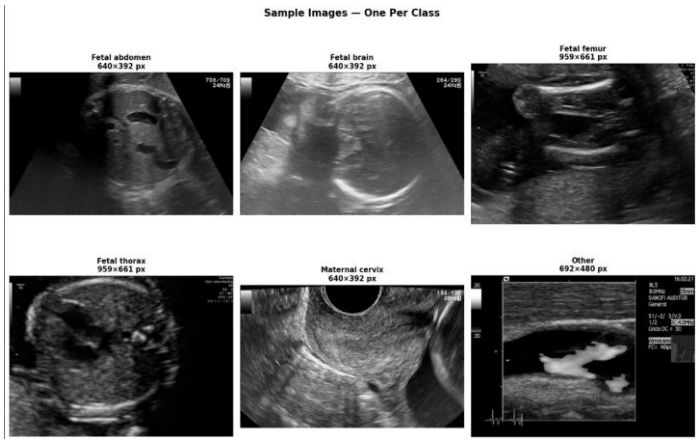

Fig. 3 Sample ultrasound images from each class.

5.2 Train / Val / Test Split

An 80 / 10 / 10 stratified split was applied at the image level.This yielded approximately 9,920 training, 1,240 validation, and 1,240 test images while maintaining class proportions across splits.

5.3 Preprocessing & Augmentation

All images were resized and normalised using ImageNet statistics (μ = [0.485, 0.456, 0.406], σ = [0.229, 0.224, 0.225]). During training, an augmentation pipeline was applied to improve robustness and generalisation.

- Geometric: Images were first resized to 280×280 and randomly cropped to 260×260, followed by random horizontal (p=0.5) and vertical flips (p=0.2) to simulate probe orientation variability.
- Spatial transformations: Random rotation (±15°) and affine transformations including translation (±10%) and scaling (0.9–1.1) were applied to account for variation in probe positioning.
- Intensity: Colour jitter (brightness ±30%, contrast ±30%, saturation ±20%, hue ±0.05) was used to simulate device-specific imaging variations.
- Noise simulation: Gaussian blur ($\sigma \in [0.1, 1.5]$) was applied to mimic ultrasound speckle smoothing.
- Regularisation: Random erasing (p = 0.2) was used to improve robustness to occlusions and artefacts.

For validation and testing, only deterministic resizing to 260×260 followed by normalisation was applied.

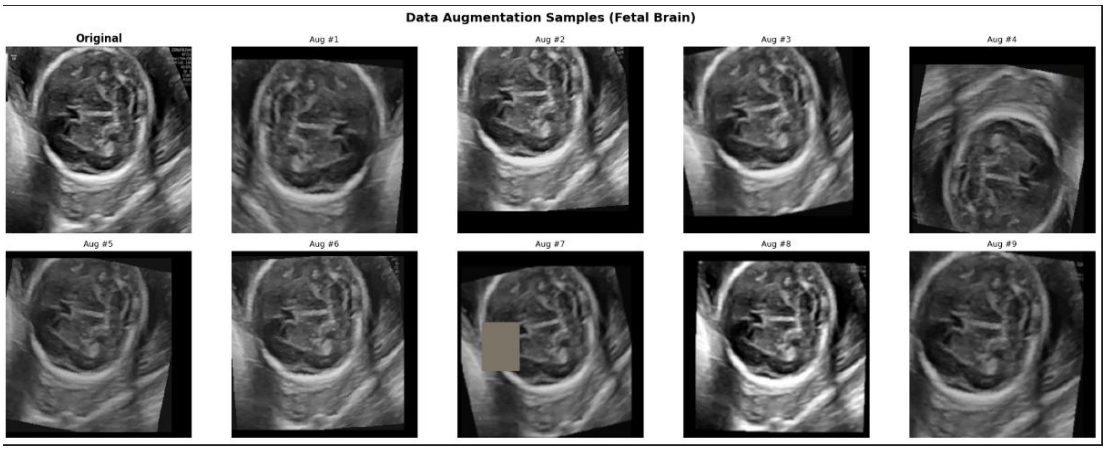

Fig. 4 Data augmentation applied during training.

6. CHALLENGES FACED

6.1 Class Imbalance

The dataset presents notable class imbalance, with the "Other" class (4,213 images) significantly larger than classes such as fetal abdomen (711 images). The proposed model effectively handles this through data augmentation and label smoothing ($\varepsilon = 0.1$), enabling stable learning and balanced performance across both majority and minority classes.

6.2 Heterogeneous "Other" Class

The "Other" class consists of diverse and non-standard ultrasound views, making it inherently complex. Despite this variability, the model demonstrates strong robustness, maintaining reliable predictions across this heterogeneous category, as reflected in the confusion matrix.

6.3 Intra-class Variation in Fetal Brain Scans

The fetal brain class (3,092 images) includes multiple anatomical sub-planes, introducing significant intra-class variation. The model successfully captures these variations, achieving high classification performance by learning generalized and discriminative feature representations.

7.EXPERIMENTAL RESULTS AND EVALUATION:

7.1 Quantitative Performance

The model achieves a weighted F1-score of **0.9218** on the validation set. High performance is observed for **fetal brain (0.9855)** and **maternal cervix (0.9909)**, indicating strong class separability. However, reduced recall for the **"Other" class (0.8314)** reflects its heterogeneous nature and overlap with multiple anatomical categories. Similarly, relatively lower precision in classes such as **fetal abdomen and thorax** indicates minor inter-class confusion.

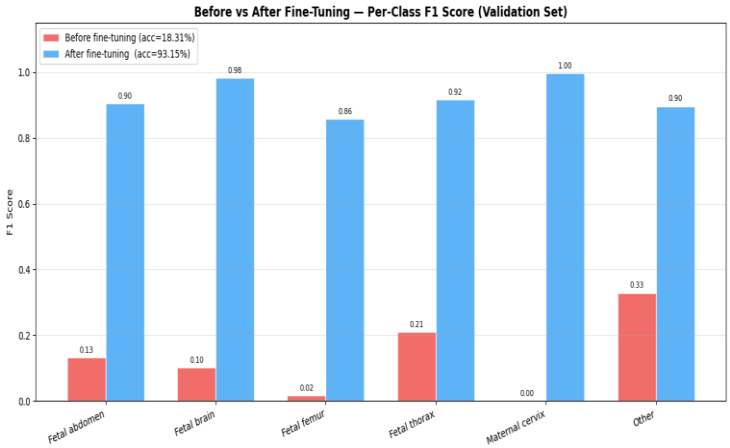

Fig. 5 Per-class F1-score before vs after fine-tuning.

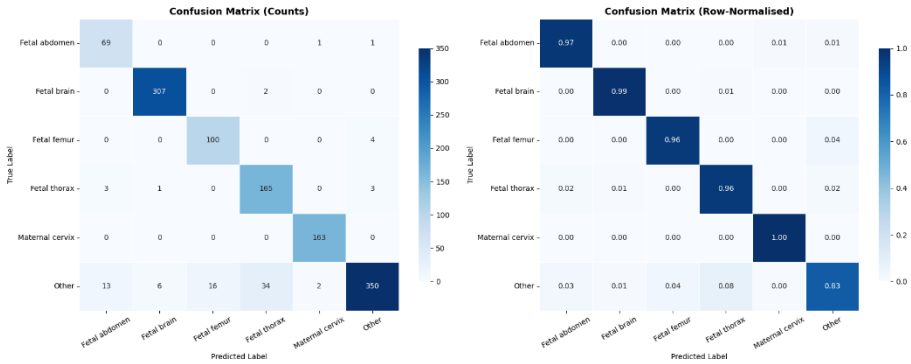

Fig. 6 Confusion matrices (counts and normalized) of EfficientNet-B3.

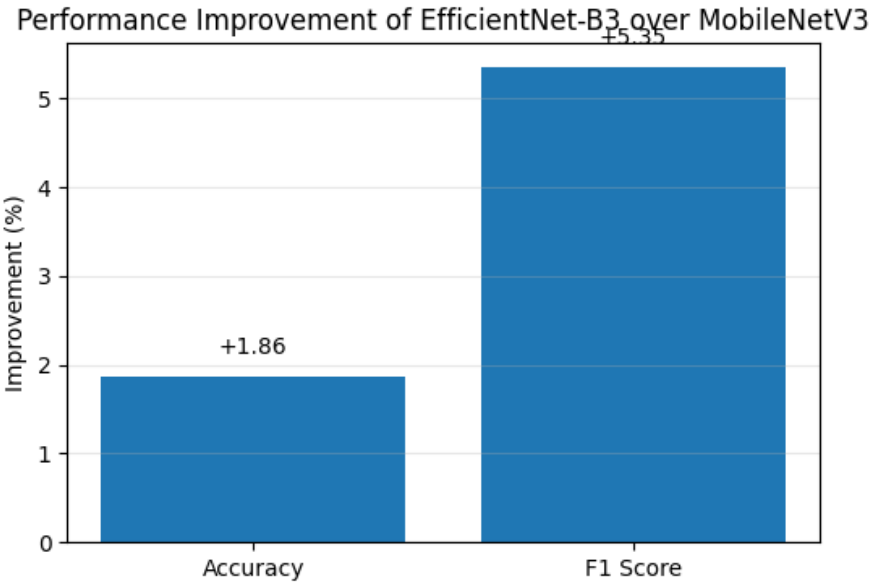

Fig. 7 Improvement of EfficientNet-B3 over MobileNetV3**.**

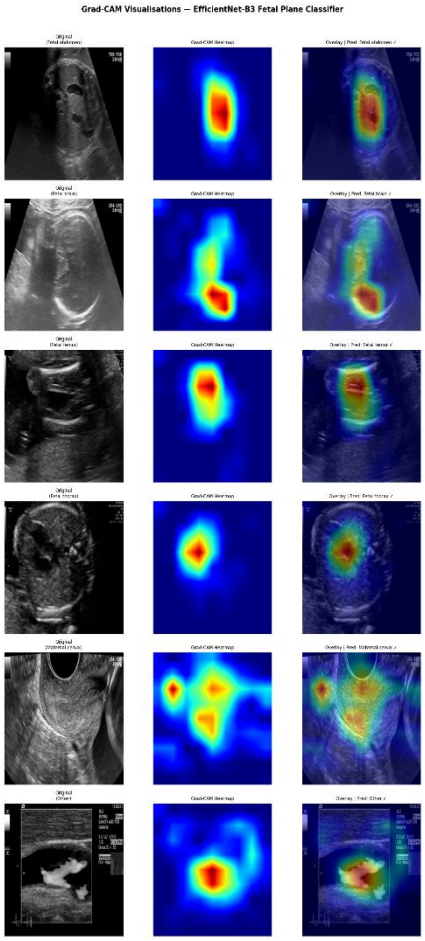

Fig. 8 Grad-CAM visualizations

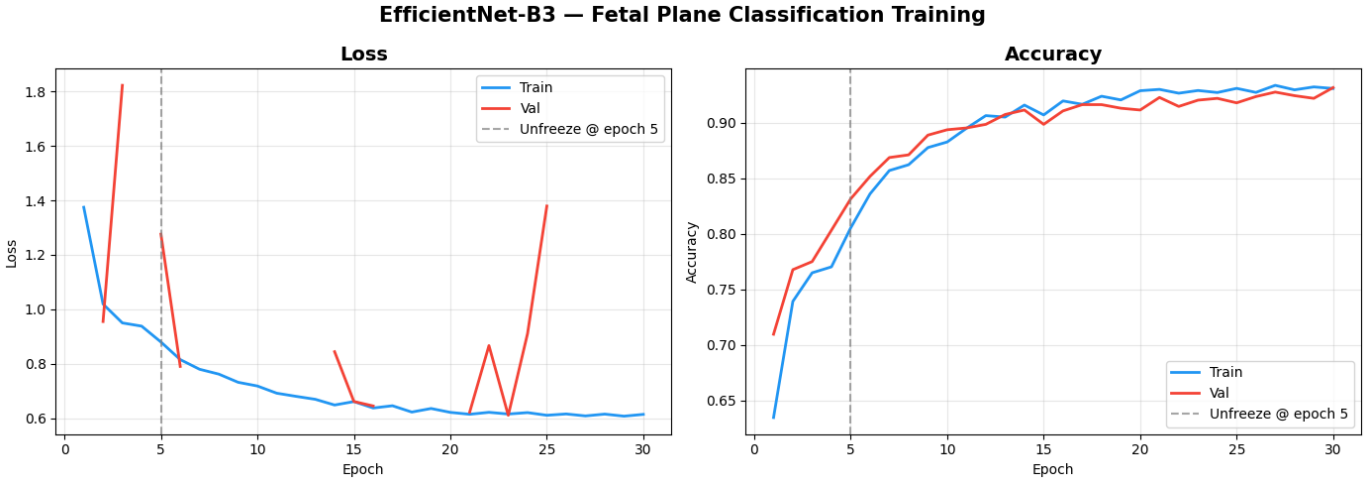

Fig.9 Training and validation loss and accuracy curves (EfficientNet-B3).

7.2 Training Dynamics

Training converged steadily within **22 epochs**, with early stopping applied (patience = 8). The transition at **epoch 5**, corresponding to backbone unfreezing, introduces a temporary increase in validation loss followed by rapid stabilization and improvement. This confirms the effectiveness of the **two-stage fine-tuning strategy**, as direct end-to-end training resulted in unstable convergence in preliminary experiments.

7.3 Qualitative Analysis (Grad-CAM)

Grad-CAM visualizations confirm that the model focuses on clinically relevant anatomical structures. For instance:

- Fetal brain → activation around cranial boundaries
- Fetal femur → emphasis on bone shaft (diaphysis)
- Fetal abdomen → central abdominal region

For challenging classes such as maternal cervix and "Other", attention maps are more diffuse, reflecting ambiguity and variability in these categories. This explains the comparatively lower performance in these classes.

7.4 Model Comparison

To assess the effectiveness of the proposed approach, **EfficientNet-B3** was compared with **MobileNetV3-Large**, a lightweight convolutional neural network designed for computational efficiency. MobileNetV3 achieved a test accuracy of **91.29%** with a weighted F1-score of **0.9145**, demonstrating strong performance with reduced model complexity. However, comparatively lower precision and recall were observed in challenging and heterogeneous classes.

In contrast, EfficientNet-B3 achieved a higher test accuracy of **93.15%** and a weighted F1-score of **0.9680**, indicating improved class-wise balance and robustness. The performance gains can be attributed to EfficientNet's compound scaling strategy, which enables more effective feature extraction across varying anatomical structures. Overall, while MobileNetV3 provides an efficient baseline, EfficientNet-B3 demonstrates superior performance, making it more suitable for accurate fetal ultrasound plane classification.

8. CONCLUSION, KEY TAKEAWAYS AND FUTURE DIRECTIONS:

This work shows that EfficientNet-B3, combined with transfer learning and progressive fine-tuning, achieves strong performance in fetal ultrasound plane classification, learning robust and clinically relevant features.

Key Takeaways: EfficientNet-B3 provides an effective accuracy–efficiency balance. Progressive unfreezing improves training stability, while augmentation and label smoothing enhance generalization. Grad-CAM analysis confirms attention to clinically meaningful regions.

Future Directions: Future work will focus on imbalance-aware learning, patient-level data splitting, and hierarchical classification strategies, along with integrating image quality assessment and explainability for reliable clinical deployment.

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GITHUB LINK : https://github.com/agi1014/Fetal-Ultrasound